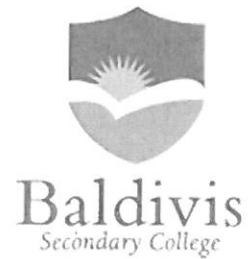


Year 9 Mini Test



Name: Answers.

Class: _____

Time: 30 minutes

Total Mark: ____/25

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Question 1

2 marks

Simplify these ratios:

i) $15 : 5$

$3 : 1$ ✓

ii) $1.8 : 1.5$

$18 : 15$

$6 : 5$ ✓

← $\frac{1}{2}$ mark

Question 2

1 marks

A mixed netball team has 4 boys and 7 girls. What is the ratio of boys to girls?

$4 : 7$ ✓

Question 3

2 marks

Express as a ratio and simplify:

i) 300g to 1000g

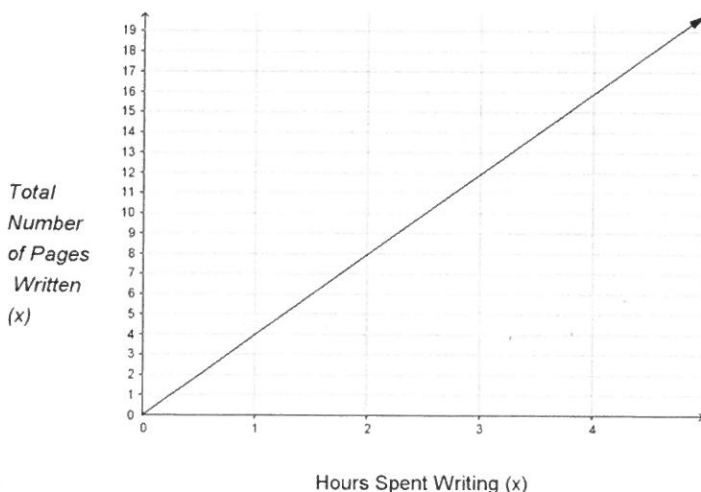
$3 : 10$ ✓

ii) 1.4m to 70cm

$2 : 1$ ✓

Question 4

1 marks



Look at the graph on the left. The total number of pages written is proportional to the number of hours spent writing (circle an answer):

i) True

ii) False

Question 5

2 marks

Which of the following pairs of ratios are in proportion?

i) $6:3$ and $12:4$ Yes. ✓

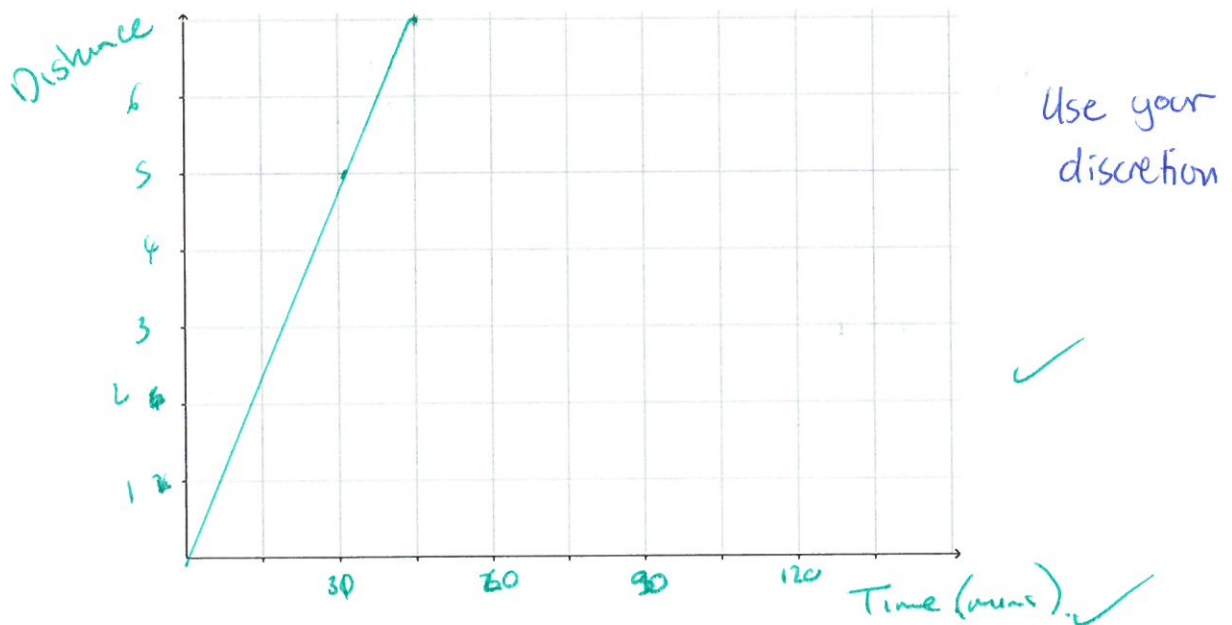
ii) 90cm at 8 years old and 1.45m at 12 years old: No. ✓
 $90:8$ $145:12$
 $45:4$

Question 6

4 marks

Rodrigo can walk 5km in 30 minutes:

i) Draw a graph to represent this information.



ii) Write a rule for Rodrigo's journey.

$$D = \frac{1}{6} T. \quad \text{or} \quad T = 6D. \quad \checkmark$$

Question 7

4 marks

Assuming direct proportionality. If \$AU1 can be exchanged for \$US1.09:

i) How many Australian dollars can be exchanged for \$US500?

$\$458.72$ $\$459.$ ✓ ✓

ii) How many US dollars can be exchanged for \$AU900?

$\$981$ ✓ ✓

Question 8**2 marks**

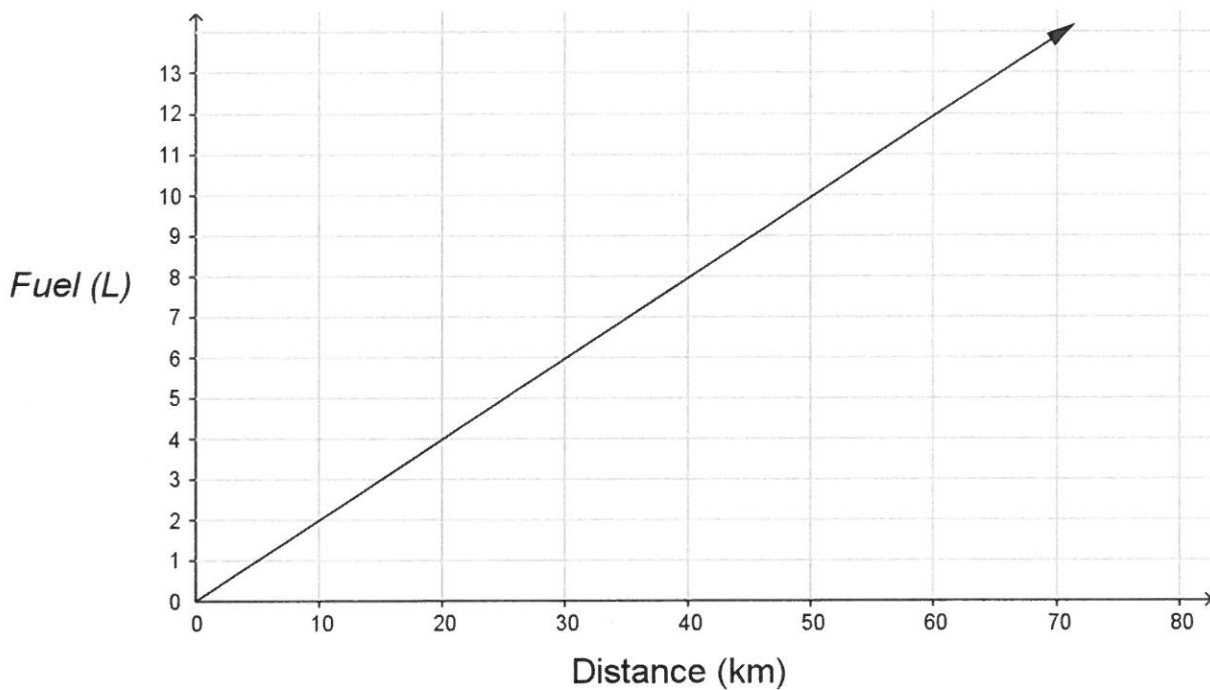
If 8 people can build a house in 20 days, assuming inverse proportionality, how long would it take 20 people to build the same house?

~~20~~ 8

✓ ✓

Question 9**5 marks**

The following graph represents the fuel consumption of a regular truck:



- i) How far can the truck go on 10L of fuel?

50km.

✓

- ii) How much fuel is required to go 1km?

0.2L

✓

- iii) What is the rule?

$$F = \frac{1}{5} D.$$

✓

✓

- iv) Using your rule how much fuel would be needed to travel 200km?

40L.

✓

Question 10**2 marks**

If 1 kilogram of bananas cost \$4.00:

- i) What is the cost of 6kg?

\$24



- ii) How much bananas for \$9.25?

2.3kg.



END OF TEST